Prediction Modeling of Crypto System Vulnerabilities

Progress Update - March 2026

Eileen R & Atharv K

Quick Recap of Project

1 **Ultimate Focus**

Determine a possible relationship/correlation between current events/variables to data breaches occurring within firms to prevent devaluation and bankruptcy of firm.

2 **Project Aim**

Analyze the relationship in specific variables: societal, economic, geopolitical, and technological to data breaches in firms.

3 **Different Approach**

Initial approaches focused on calculating specific breaches. Our project simply aims to figure out the probability of the breach occurring based on historical attempts and external catalysts.

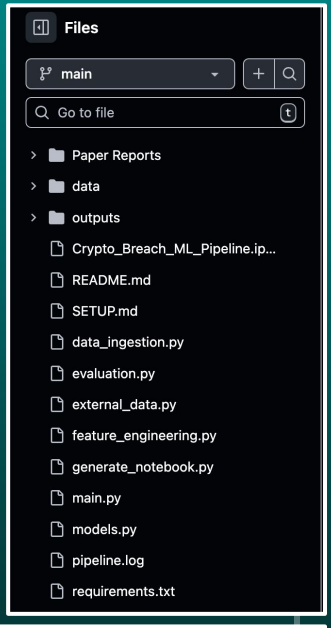
4 **Why?**

Provide a way for companies to be prepared for potential external threats before they occur.

Current Progress To Date

1 Code Setup

- Utilizing Python as main language.
- 3 Main Elements:
  - *Code Files (Notable):*
    - *main.py* - reads data, performs training, provides evaluation.
    - *data.ingestion.py* - loads all datas sources in into organized manner.
    - *models.py* - contains the models that will be trained.
    - *evaluation.py* - provides metrics of the model (brier score, log-loss, PR-AUC).
    - *Crypto_Breach_ML_Pipeline.ipynb* - runs the stages of the pipeline (data ingestion, feature engineering, training, risk scoring, etc).



  - */data* - contains the data to feed and train the model.
    - Data Breaches (Year, Organization, Country)
    - Market Trends
    - World Bank Data

  - */output* - graphs, matrices, and evaluation metrics from model.
    - Dataset Overview
    - Attack Method Distribution
    - PR Curves
    - Brier & Log Loss

1		Number	Name_of_Covered_Entity	State	Business_Associate_Involved	In
2	1	0	Brooke Army Medical Center	TX		1
3	2	1	Mid America Kidney Stone Association, LLC	MO		1
4	3	2	Alaska Department of Health and Social Services	AK		5
5	4	3	Health Services for Children with Special Needs, Inc.	DC		3
6	5	4	L. Douglas Carlson, M.D.	CA		5
7	6	5	David I. Cohen, MD	CA		8
8	7	6	Michele Del Vicario, MD	CA		6
9	8	7	Joseph F. Lopez, MD	CA		9
10	9	8	Mark D. Lurie, MD	CA		5
11	10	9	City of Hope National Medical Center	CA		5
12	11	10	The Children's Hospital of Philadelphia	PA		9
13	12	11	Cogent Healthcare of Wisconsin, S.C.	TN		6

Current Progress To Date Pt 2

2 Data Collection

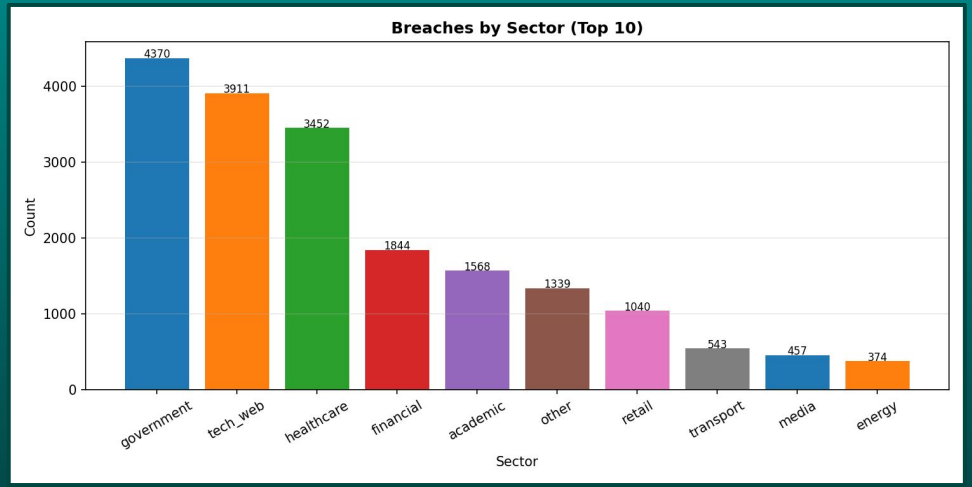
- **8** target datasets have been placed into a single, unified schema.
- **19,281** different breach records from 2004 - 2026.
  - Cyber Events Database & Washington State Breach Notifications.

3 Feature Engineering

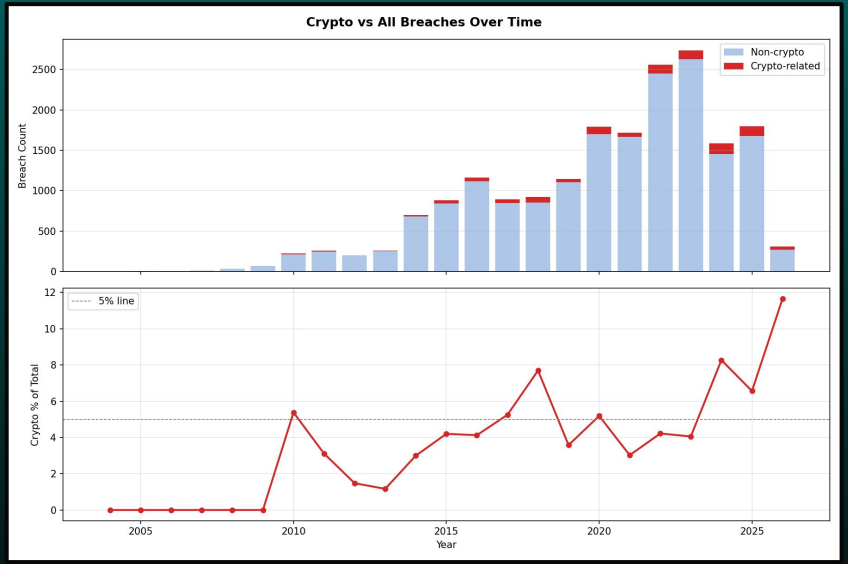
- **23**-row annual feature matrix with **65 distinct features** has been created via the data breaches.
- Main feature groups:
  - Total Breaches, Log-Records
  - Sector and Method Fractions
  - Year-over-Year trend signals

4 Model Training and Evaluation

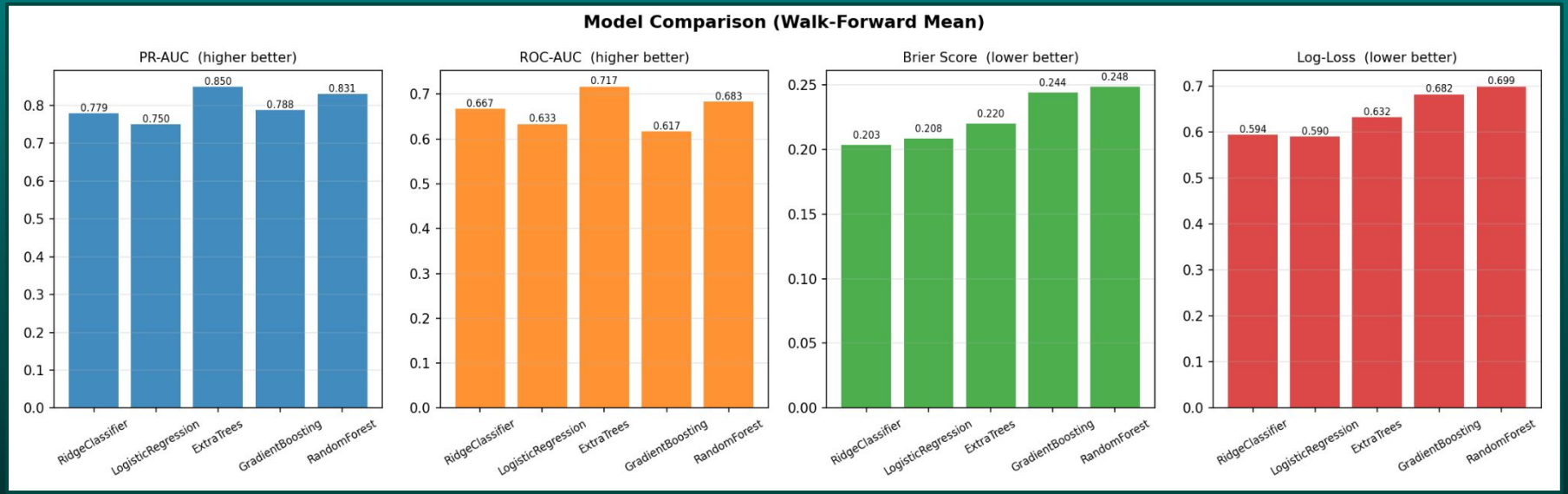
- **5 Pipelines** trained and evaluated across 17 walk-forward folds.
  - Logistic Regression (Classification Metric).
  - Ridge Class (Similar to LR, but more strict with outliers).
  - Extra Trees (Tree splits based on feature values).
  - Random Forest (Collection of “extra trees”).
  - Gradient Boost (Builds the trees sequentially).
- Ran the model on years **2009 - 2026**.



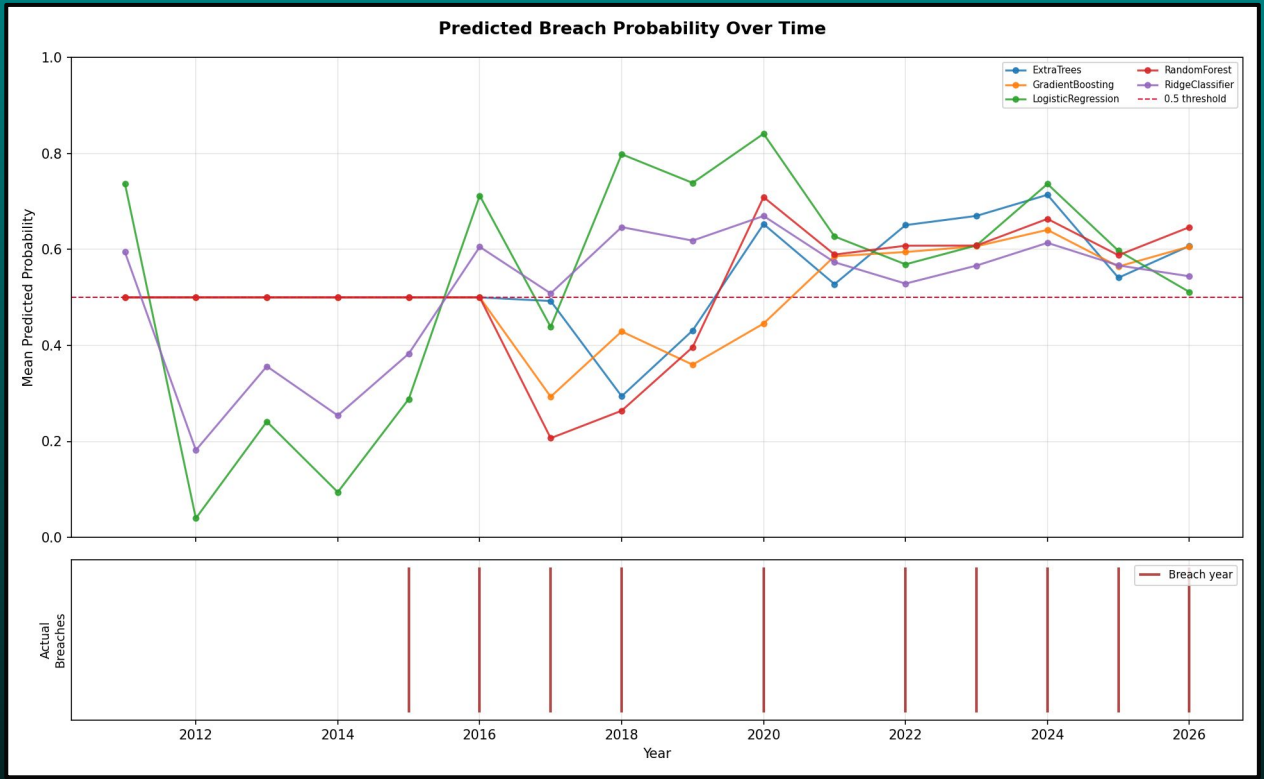
Government and Tech are top 2 breach sectors (2).



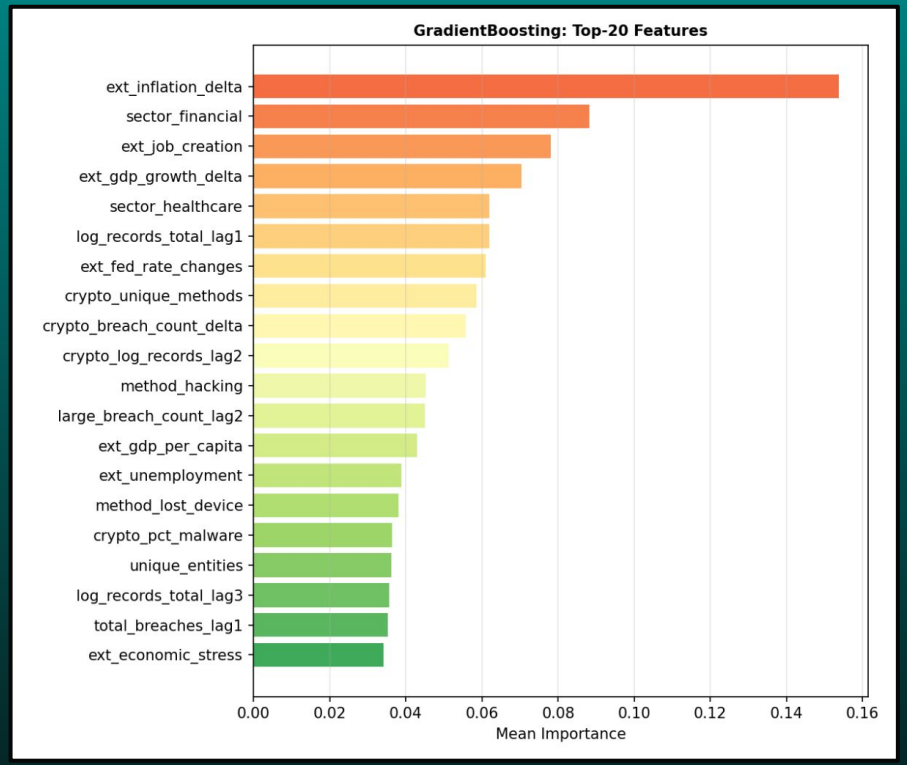
Crypto Growth Over Time (4)



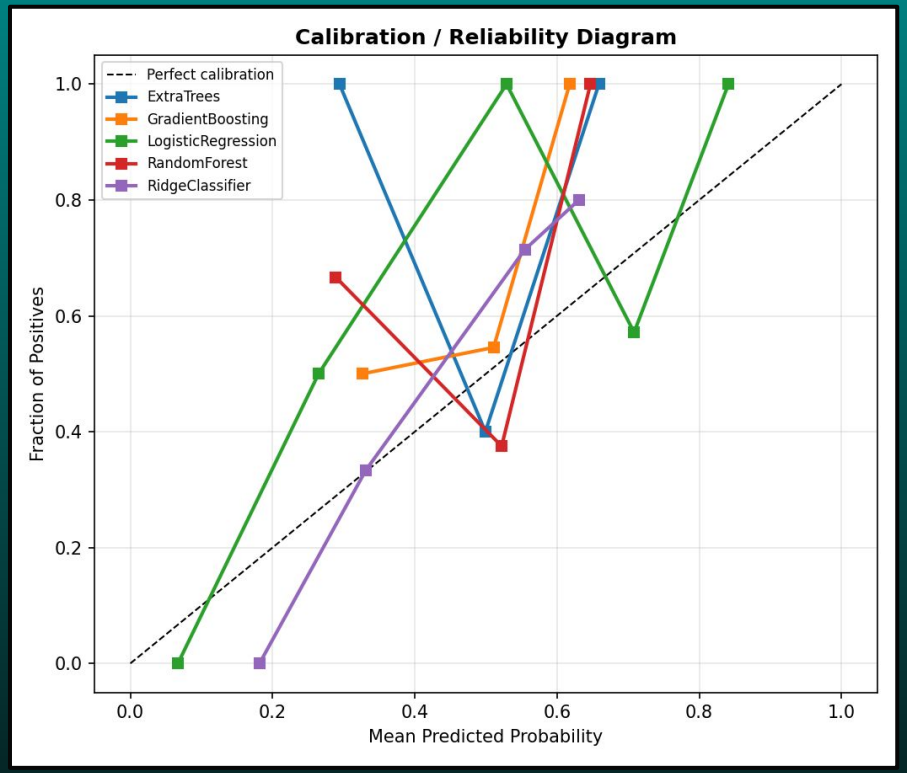
Model Comparison with Different Metrics (6)



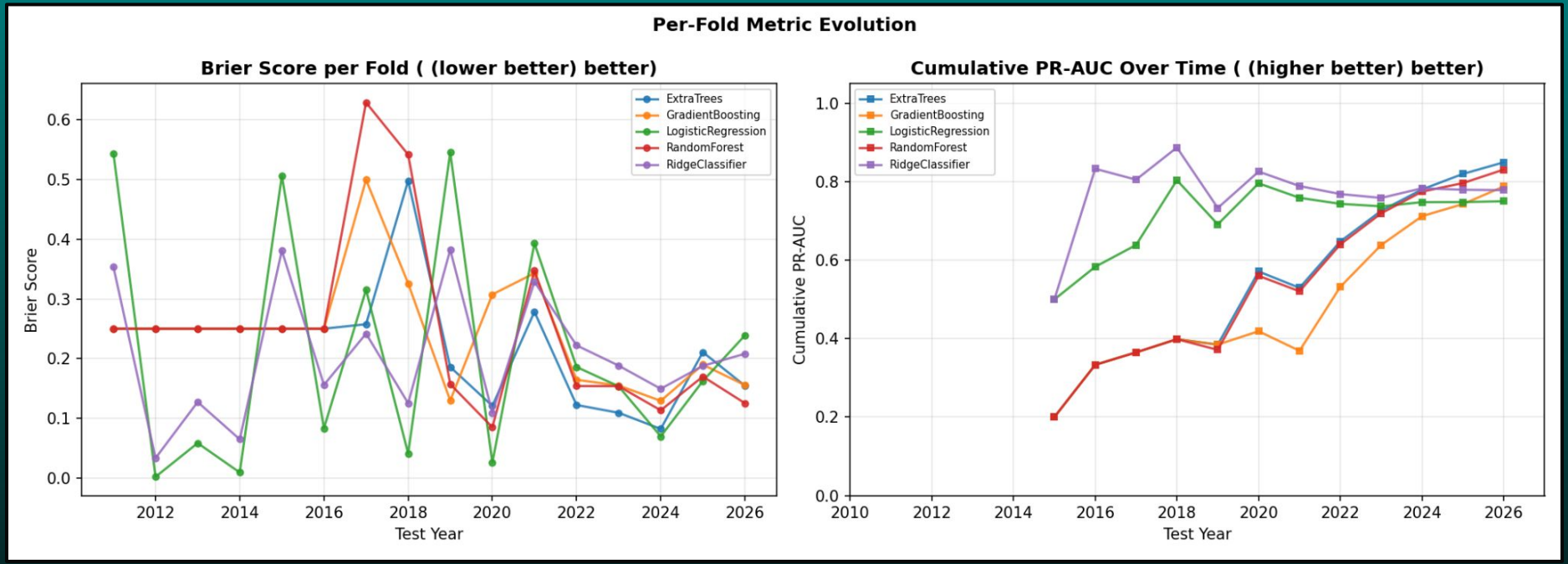
Breach Probability Calculation Over Time (8)



Breach Probability Calculation Over Time (9)



Calibration/Reliability Diagram with Models (10)



Brier Score & Cumulative PR-AUC per Fold (12)

Key Findings From Project Thus Far

Key Find 1

Crypto fraction grew from **0%** to around a peak of **12%** (2004 - 2026).

Key Find 2

Affect on crypto came from **macroeconomic events, not breach-specific.**

Events that represented financial instability caused the most damage (VIX, economic stress, current account balance, etc).

Key Find 3

Hacking, **Malware,** and **Phishing** are incredibly common attack methods.

Hacking: **7,253** incidents
Malware: **5,293** incidents
Phishing: **2,038** incidents

Key Find 4

Logistic regression was the best method out of the 5 models.

Outperformed tree models in **small N-time series.**

MODEL COMPARISONS

Model	Brier	LL	PR	ROC
Logistic Reg.	**0.088**	**0.322**	**0.991**	**0.900**
Ridge Class.	0.139	0.461	0.991	0.900
Extra Trees	0.154	0.485	0.958	0.600
Random Forest	0.155	0.489	0.971	0.700
Grad. Boost.	0.194	0.577	0.944	0.500

0.088
Brier Score

0.322
Log Loss

0.991
PR

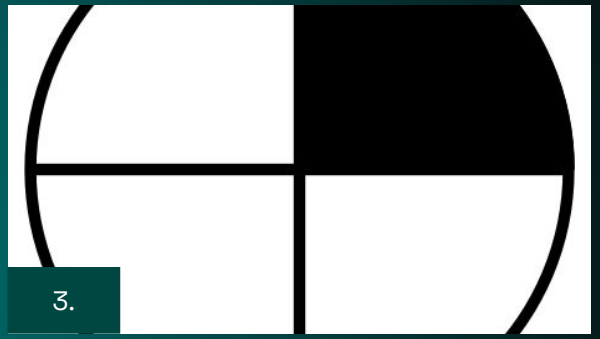
Issues and Future Focuses

Scale Imbalance

- Adding more data lead to an imbalance between the high vs low thresholds.
- Utilized a percentage to represent the crypto breaches instead (led to improved balance).
    - # of crypto breaches / total breaches
- **Concern:** Total breaches may not be independent.

Overconfidence

- Applicable to the low-probability regions.
- Overestimating certainty which means it's never predicting low probabilities.
- Dataset seems to be imbalance towards positive (HIGH).
- **Concern:** Want to make probabilities more honest and less overconfident in cases where datasets are small.



Partial 2026 Data

- Dataset only contains 3 months worth of data breaches for 2026.
- 2026 appears to be a low year, but could not be.
    - All models are predicting high probability of breaches for 2026.
- **Concern:** Dealing with incomplete data for a certain time series.

Thank you!

Questions?